

INTRODUCTION

Preserving fragile history as interactive replicas

Historical artifacts — papyrus manuscripts, cuneiform tablets — are fragile, rare, and often kept out of reach of the people who want to study them.

Digital Twins of Artifacts builds accurate, interactive, web-viewable 3D replicas so these objects can be examined, taught, and displayed without handling the original. It began as RIT's **Freshman Imaging Project**, grew into **Extended-FIP (EFIP)**, and runs with the **RIT Cary Graphic Arts Collection**, which provides the artifacts.

APPROACH

Two purpose-built imaging systems

Each class of artifact is paired with a rig designed for it. Which one an object uses depends on a single question: **does its shape carry meaning in three dimensions?**

- **Flat, detail-rich** objects (papyrus, parchment) → a 2D photometric-stereo rig that recovers how the surface responds to light.
- **Truly three-dimensional** objects (cuneiform tablets, seals) → a 3D photogrammetry rig that reconstructs real geometry.

SYSTEM A · PAPYRUS & FLAT MANUSCRIPTS

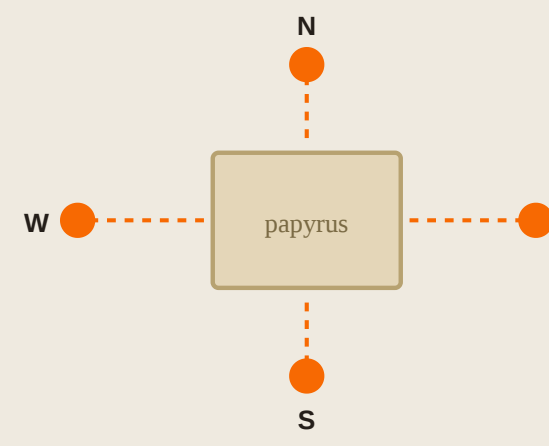
2D Photometric-Stereo Rig

A Canon DSLR paired with an Arduino-driven directional lighting array photographs the manuscript once per light direction. Instead of reconstructing geometry, it recovers how the surface *responds to light* — revealing texture, ink, and fine relief.



Photo — 2D papyrus capture rig

DSLR + directional lighting array · landscape ~4:3



Four directional lights (N/E/S/W) at ~37° elevation. A stepper-rotated polarizer captures each direction as **cross-polarized** (clean geometry) and **co-polarized** (reveals ink) pairs.

SYSTEM B · CUNEIFORM TABLETS & OBJECTS

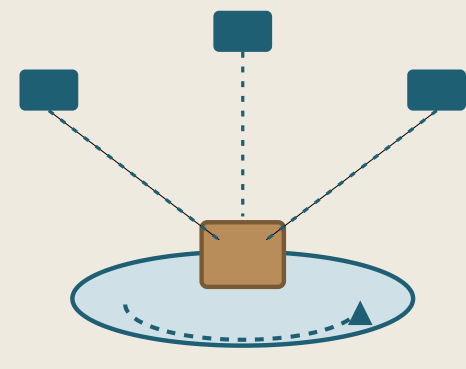
3D Photogrammetry Rig

A multi-camera rig on an Arduino-driven turntable photographs the artifact from many angles. GPU-accelerated structure-from-motion and dense stereo build a point cloud and, ultimately, a watertight 3D mesh — a true replica of the object's form.



Photo — 3D tablet capture rig

Multi-camera + turntable · landscape ~4:3



Multiple cameras shoot the object as the **turntable rotates**. Both sides are captured, then merged into one model.

METHODS

From raw captures to a digital twin

PAPYRUS Photometric-stereo pipeline (Python)

0

Segment

ML segmentation isolates the fragment from the background.

1

Calibrate

Flat copy-paper shots remove vignetting & light falloff.

2

Core maps

Normal, albedo, specular & roughness from cross/co-pol pairs.

3

Height

Surface normals integrated via weighted Frankot–Chellappa.

4

Bake

Maps baked into a three.js GLB model for the viewer.

TABLETS Photogrammetry pipeline (COLMAP / Open3D)

1

Clean

Background removed from every photograph.

2

Reconstruct

CUDA COLMAP structure-from-motion + dense stereo → point cloud.

3

Align

FPFH + RANSAC, refined with ICP, merges both sides.

4

Mesh

Open3D Poisson reconstruction → watertight textured model.

RESULTS

Outputs



Papyrus texture maps

normal / albedo / specular / height

Calibrated surface maps. The building blocks of the papyrus twin, recovered from the polarized-light captures.



Cuneiform-tablet twin

reconstructed 3D mesh

Watertight 3D replica. Both faces of the tablet reconstructed and merged into one model.



Interactive web viewer

in-browser inspection

Explore in the browser. Rotate, relight, and inspect each twin — no special software needed.

ACCESS

An interactive web viewer

Every finished twin lives in a static three.js gallery that runs in any browser. Visitors can rotate each artifact, adjust the lighting, and swap the backdrop to study the surface under different conditions.

- **Hand-tracked magnifier** — a webcam tracks the visitor's fingertip and a magnifying lens follows it across the artifact, like a museum exhibit.
- **Adjustable relighting** reveals ink and relief that a single fixed view would hide.
- **Runs anywhere** — a static site with no installs or plugins, so the collection can be shared with a single link.

REFERENCE

Technical specifications

Camera

Canon DSLR, tethered; RAW → 16-bit TIFF (~5208 × 3476 px)

Lighting

4 directional lights (N/E/S/W) at $\theta \approx 37^\circ$; rotating polarizer

Polarization

cross-pol (geometry) + co-pol (ink) capture pairs

Height

weighted Frankot–Chellappa integration of surface normals

3D reconstruction

CUDA COLMAP (SfM + MVS) → FPFH/RANSAC + ICP → Open3D Poisson mesh

Rendering

three.js physical material baked to GLB; static viewer w/ MediaPipe hand tracking

CONCLUSION & FUTURE WORK

A repeatable path to digital preservation

Two complementary rigs and their automated pipelines turn fragile originals into faithful, shareable digital twins — a workflow that repeats across artifact types and delivers results anyone can explore in a browser.

- Expand the collection of scanned artifacts with the Cary Collection.
- Refine capture speed and automation for both rigs.
- Grow the public web gallery and its interactive features.

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